Essays

|!|BCEEB0230000001|!| ★ **STSE**

Since the 1950s, the number of deaths due to infectious diseases has been in decline in developed countries. Explain such decline with regard to improvements on disease prevention and treatment. (12 marks)

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Prevention of disease: (Max. 5)

- Doctors and scientists have gained deeper understanding about the transmission routes of pathogens. (1) This allows people to take measures to break the transmission link and thus control the spread of the pathogens. (1)
- Improved personal hygiene, proper treatment of drinking water and proper food processing and storage have reduced the risk of food and water contaminated with pathogens. (1)
- Proper sewage treatment has prevented the spread of pathogens from human wastes to the environment / water systems. (1)
- Improved environmental hygiene helps control the breeding of disease vectors such as rats and mosquitoes. (1) This also reduces the survival of pathogens in the environment. (1)
- Immunization programmes allow young children to have developed immunity against common infectious disease such as polio and measles. (1) Thus the number of infected persons in the community is reduced and so is the chance of disease transmission. (1)

Treatment of diseases: (Max. 4)

- Different types of antibiotics and antiviral drugs have been discovered and mass produced. (1) The drugs can quickly and effectively kill the pathogens in patients, thus reducing the death rate. (1)
- Advancements in diagnostic techniques allow quick and accurate identification of the causative agents, (1) thus treatment methods can target at the concerning pathogens. (1) The recovery time is shortened and the success rate is increased. (1)
- Faster recovery can reduce the chance of disease transmission. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Applying concepts	9
Communication	3

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|!|BCEEA0230500002|!| ★★ **STSE**

Owing to medical advances, many infectious diseases are now under control. However, new threats such as multidrug-resistant 'superbugs', SARS and COVID-19 pose new challenges to the health profession and the general public. Discuss what challenges are faced by the health profession and how the general public should co-operate to prevent the emergence and spread of noval diseases. (11 marks)

##

Challenges faced by the health profession: (Max. 4)

- The indiscriminate use of antibiotics in humans and animal rearing speeds up the emergence of antibiotic resistance among pathogens. (1) Infections caused by multidrug-resistant 'superbugs' (e.g. MRSA) are difficult to treat and if not controlled properly they may cause an epidemic. (1)
- New strains of pathogens may emerge due to mutation, e.g. avian flu and swine flu. (1) People do not have immunity against these new strains. Thus many people would be infected and the transmission could be very quick. (1)
- Urbanization of formerly rural areas may bring people into contact with pathogens from wild organisms / soil. (1) These pathogens from wild organisms / soil may transmit to humans. (1)
- Global warming increases the incidence of some infectious diseases such as malaria. (1)
- Frequent transport of people and goods facilitates the spread of diseases around the world. (1)

Ways to prevent the emergence and spread of noval diseases: (Max. 4)

- When prescribed with antibiotics, patients should finish the whole course as instructed by doctors in order to kill all the pathogens in the body. (1)
- When cases of new diseases are reported, people should actively take preventive measures and practise good personal hygiene to protect themselves from infection. (1)
- Avoid contact with wild organisms. (1)
- Use insect repellents and mosquito nets to avoid mosquito bites. (1) Improve environmental hygiene to control the breeding of disease vectors (e.g. mosquitoes and rats). (1)
- If infected with new diseases, patients should see a doctor immediately and follow isolation order as advised by doctors. (1)

Communication (Max. 3)

(11 marks)

Skills assessed	Marks
Applying concepts	8
Communication	3

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|!|BCEEC0230400003|!| **STSE**

Avian influenza and malaria are threatening diseases transmitted through different ways. Describe the causes and transmission routes of these diseases. Discuss what precautionary measures can be taken to control their spread. (13 marks)

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Avian influenza: (Max. 5)

- Avian influenza may be caused by several types of influenza viruses that originate from birds, e.g. H5N1, H9N2 and H7N9. (1)
- People may catch these viruses through contact with infected birds and their droppings. (1)
- To control the spread of avian influenza, we should
  - avoid touching birds and their droppings. (1)
  - cook the meat and eggs of poultry thoroughly. (1)
  - wash our hands thoroughly with liquid soap and water after handling chickens or eggs. (1)
  - maintain good personal hygiene. Wash hands frequently. (1)
  - consult a doctor if we have any flu-like symptoms. (1)

Malaria: (Max. 5)

- Malaria is caused by several species of the protozoan *Plasmodium*. (1)
- It is transmitted by the vectors mosquitoes. (1) The mosquito first bites an infected person and takes up the protozoans. When it bites a healthy person, the protozoans are transmitted to that person. (1)
- Malaria is not transmitted from person to person. However, malaria can be transmitted through contaminated blood or blood product transfusion, organ transplant, or shared needles or syringes. (1) Malaria may also be transmitted from a mother to her foetus/newborn baby before or during delivery. (1)
- To control the spread of malaria, we should
  - take measures to avoid mosquito bite, e.g. wear light-coloured, long-sleeved clothing (1); apply effective insect repellents to exposed parts of the body and clothing (1).
  - Avoid travelling to areas where malaria is common. (1)

(13 marks)

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Skills assessed	Marks
Understanding basic concepts	10
Communication	3

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